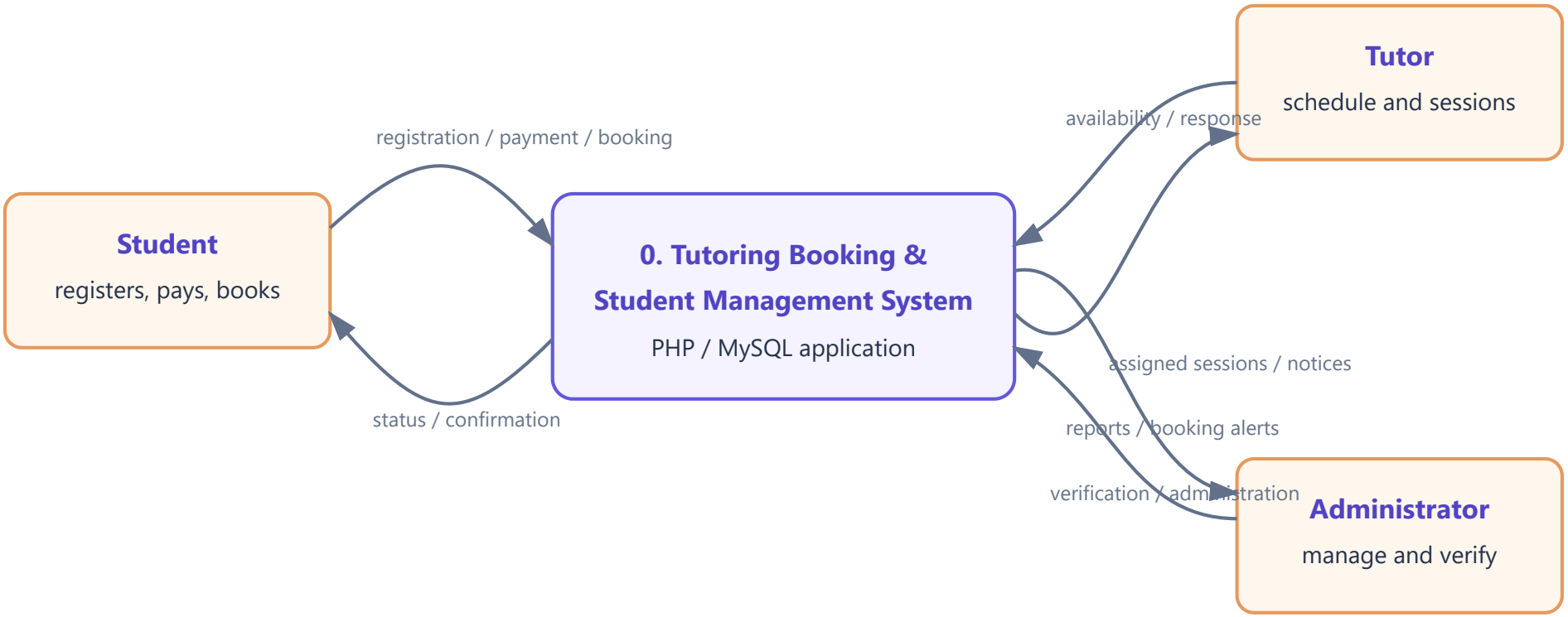
MONAGENG TUTORING

System Analysis & Design Diagrams

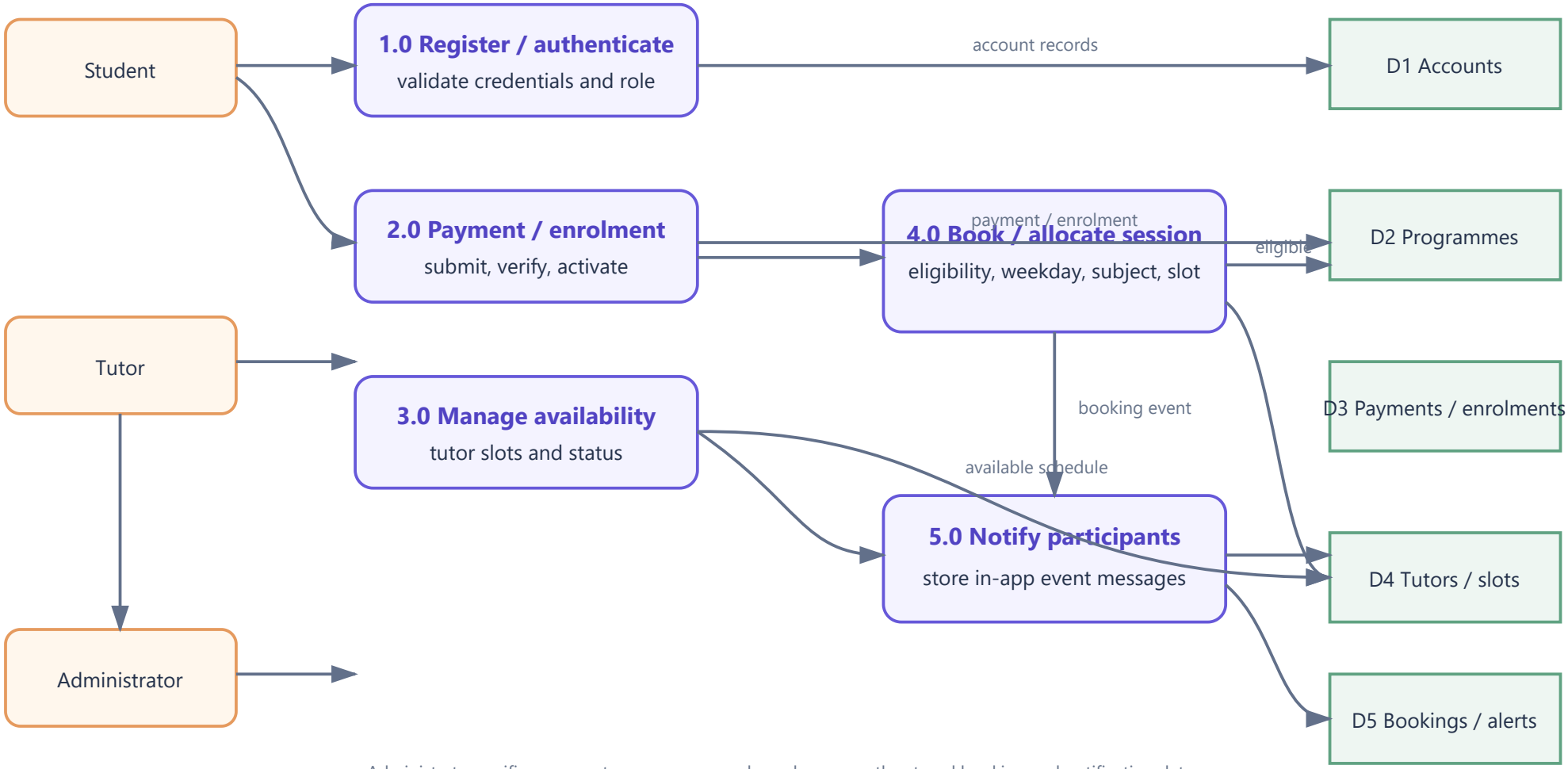
Context and Level 1 Data Flow Diagrams, role-specific Use Case Models, tutoring-session Sequence Diagram, and normalized Entity Relationship Diagram.

HPXS302-1 — Summative Assessment 1
Kgalalelo Monageng · 30HA2400106

1. DFD Context Diagram (Level 0)

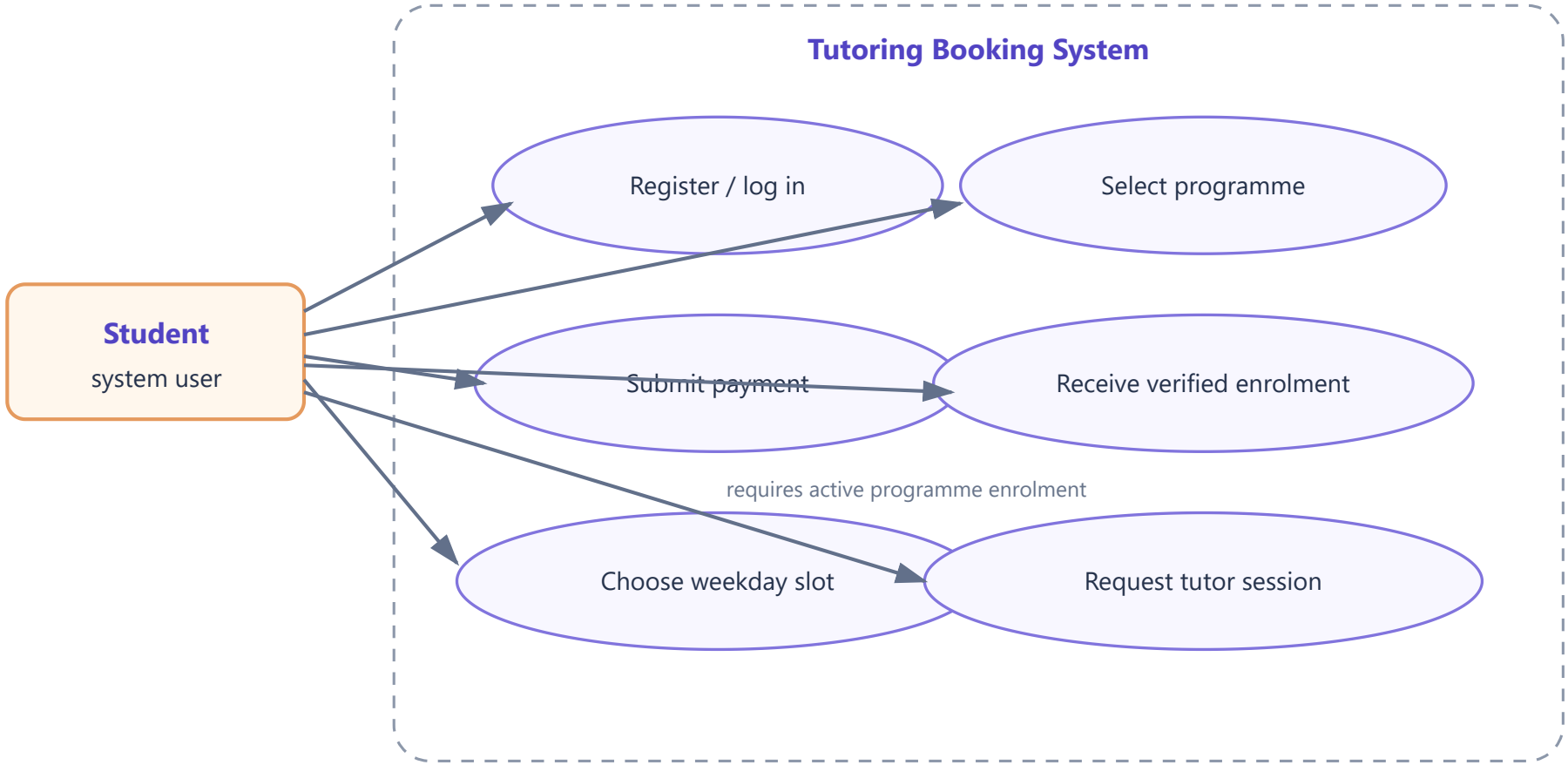


2. DFD Level 1 — Process Decomposition

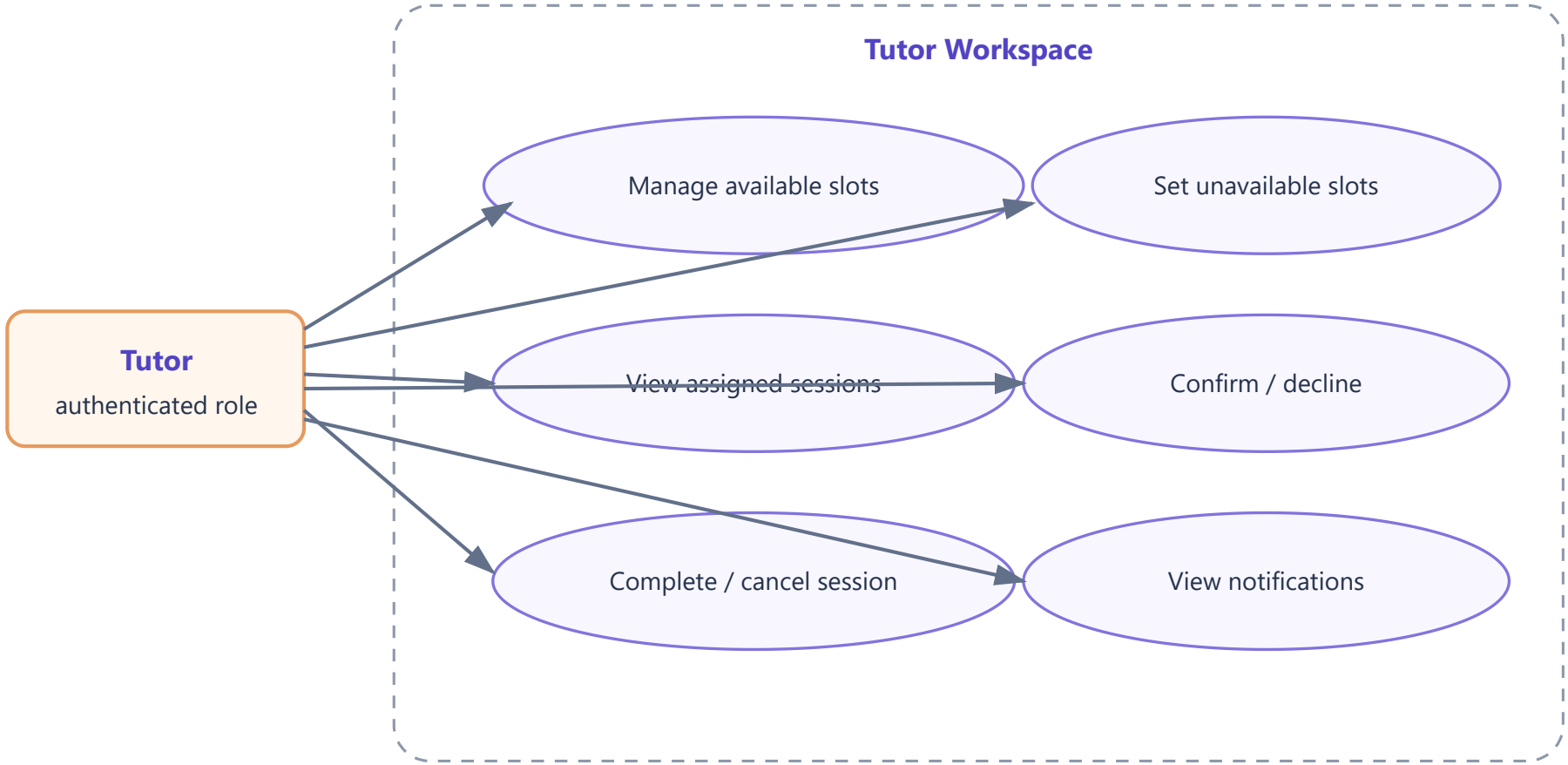


Administrator verifies payments, manages records, and oversees the stored booking and notification data.

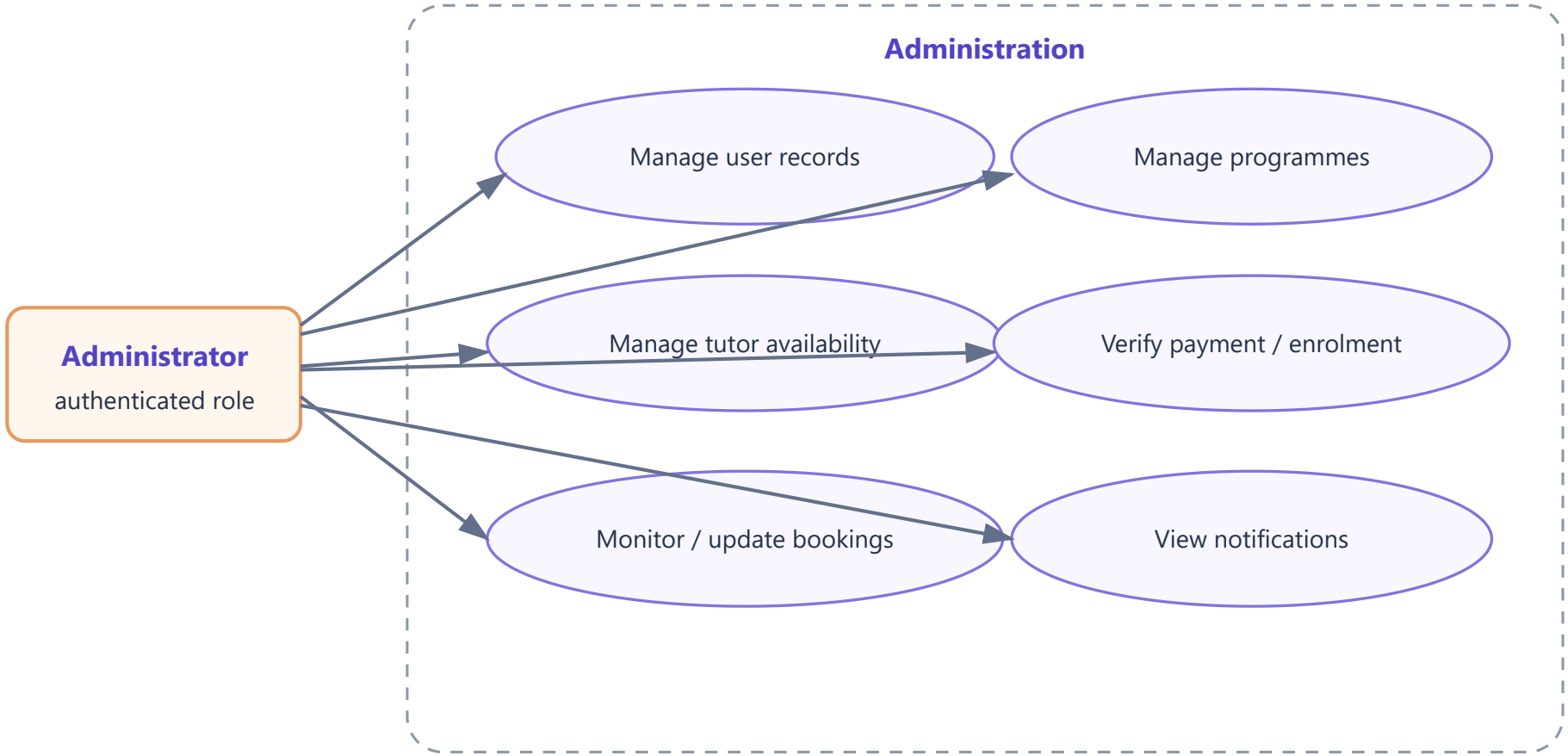
3. Use Case Model — Student Session Booking



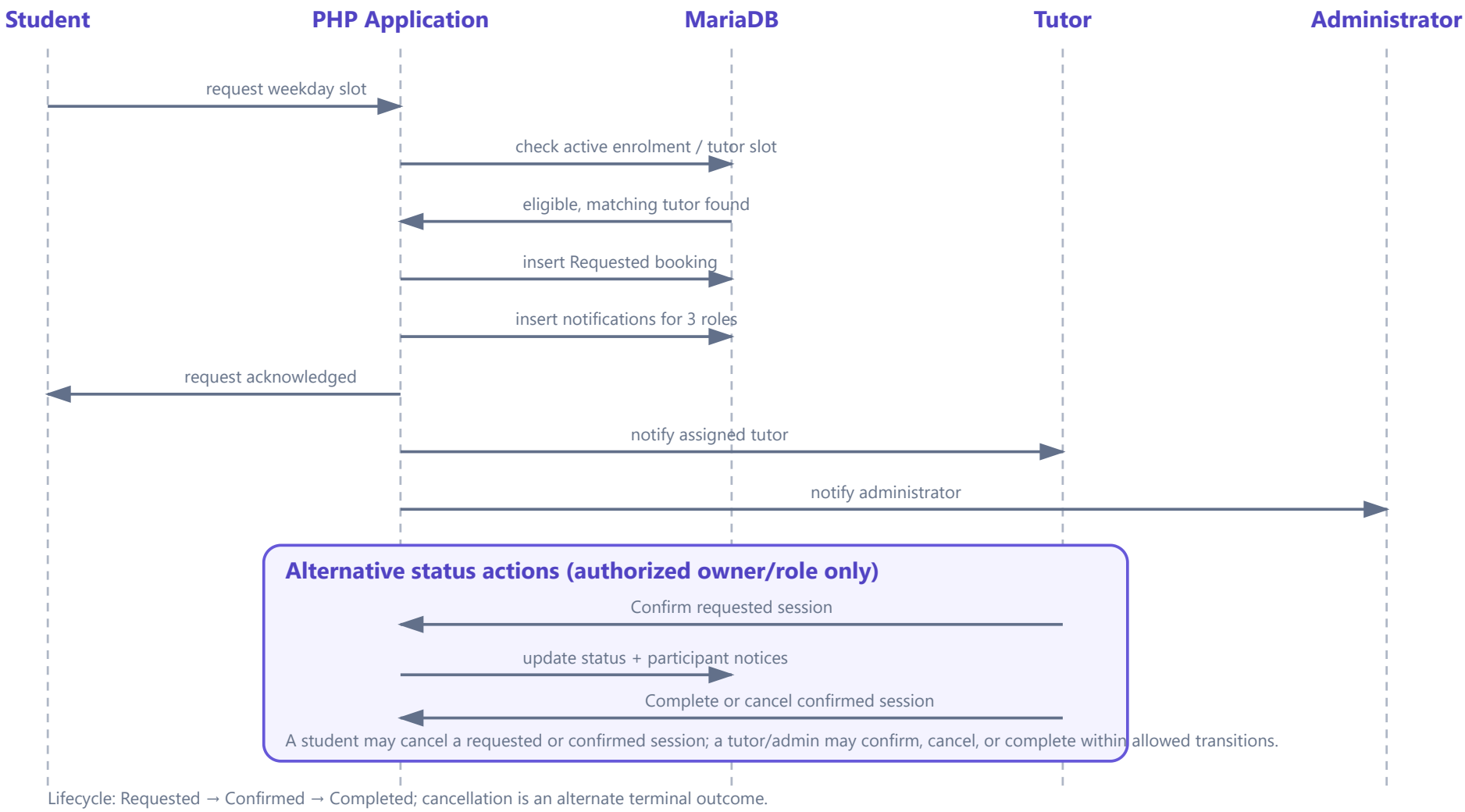
4. Use Case Model — Tutor Schedule Management



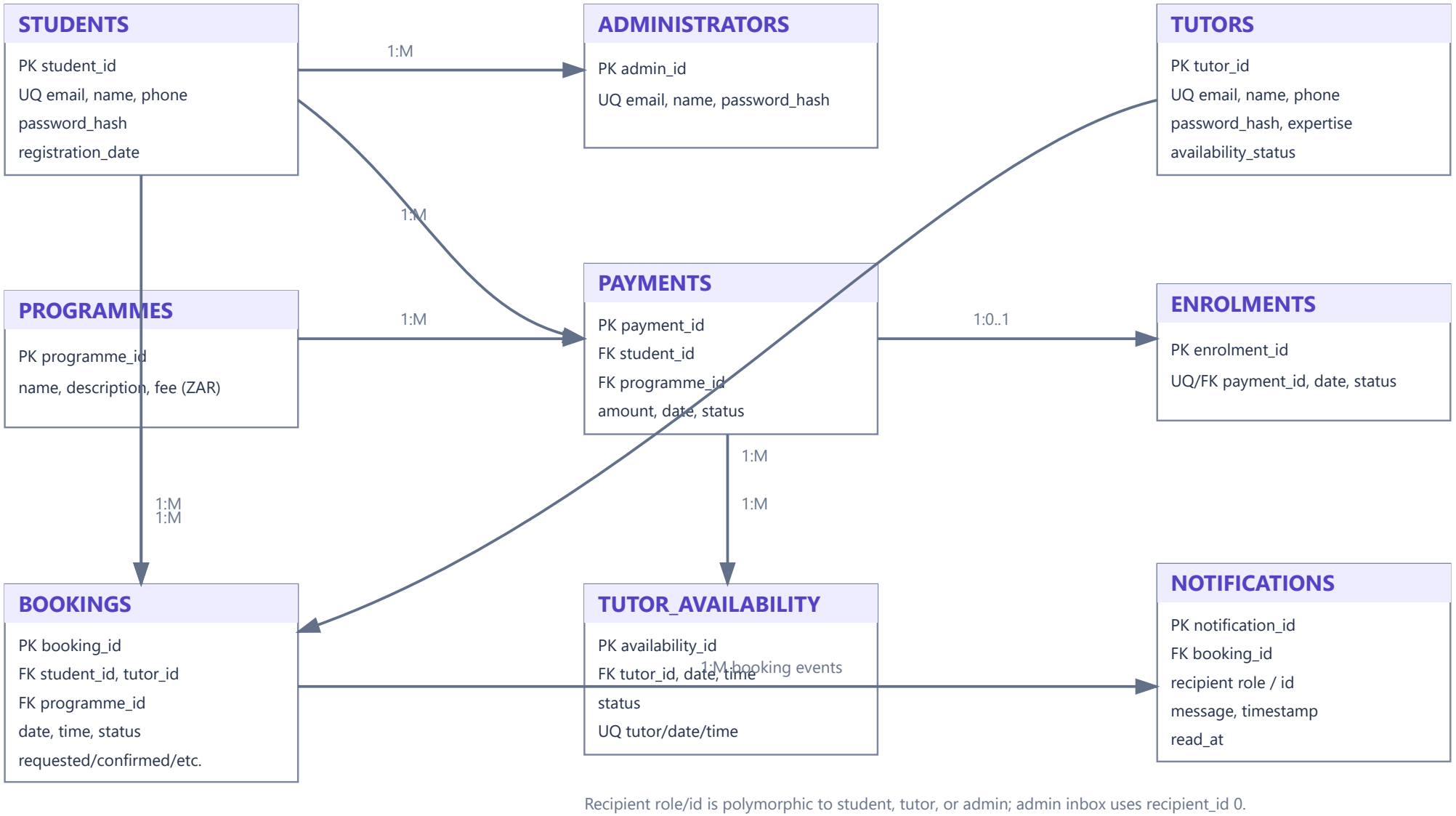
5. Use Case Model — Administrator Monitoring



6. Sequence Diagram — Tutoring Session Lifecycle



7. Entity Relationship Diagram (3NF)



Recipient role/id is polymorphic to student, tutor, or admin; admin inbox uses recipient_id 0.

Students 1:M Payments and Bookings
Programmes 1:M Payments and Bookings

Payments 1:0..1 Enrolment
Tutors 1:M Bookings and Availability

Bookings 1:M Notifications

Administrator receives role-addressed notifications and manages system records.